

MATH3303 – Lecture 23

Quotient Rings and Ring Homomorphisms

Review Notes

Reference: Lee, Abstract Algebra, Sections 9.2–9.3

Overview

This lecture builds on the notion of an ideal from Lecture 22 to construct *quotient rings*, then introduces *ring homomorphisms*:

- **Quotient ring** R/I – when $I \trianglelefteq R$, the set of cosets $\{r+I : r \in R\}$ inherits a ring structure
- **Well-definedness of multiplication on R/I** – this is what makes the absorption property of an ideal essential
- **Two-sided ideals** – needed for quotient rings when R is non-commutative
- **Ring homomorphism** – a map preserving both ring operations (and the identity)
- **Kernel and image** – $\ker \varphi$ is a two-sided ideal of R , $\text{Im } \varphi$ is a subring of S

Connection to Lecture 22. Lecture 22 introduced ideals and showed that for $I \trianglelefteq R$, the cosets $r + I$ form an abelian group under addition. Here we equip the coset space with multiplication and verify the ring axioms. The role of an ideal (compared to a normal subgroup in group theory) is precisely to make coset multiplication well-defined.

1 Quotient Rings

Throughout this section, unless stated otherwise, R is a commutative ring and I is an ideal of R .

Remark 1.1: Cosets of an ideal (from lecture)

Let R be a commutative ring and $I \trianglelefteq R$. The *left coset* of I in R with representative $r \in R$ is

$$r + I = \{r + i : i \in I\}.$$

Since $(R, +)$ is abelian, I is an additive normal subgroup, so left and right cosets coincide. We write R/I for the set of cosets, and

$$(R/I, +) \text{ forms an abelian group}$$

under $(a + I) + (b + I) = (a + b) + I$.

Definition 1.1: Quotient ring (from lecture; cf. Lee, Def. 9.3)

Let R be a commutative ring and $I \trianglelefteq R$. The *quotient ring* R/I is the set of all left cosets $\{r + I : r \in R\}$ together with the operations

$$(a + I) + (b + I) = (a + b) + I, \quad (a + I)(b + I) = ab + I,$$

for all $a, b \in R$.

Theorem 1.1: R/I is a ring (from lecture; cf. Lee, Thm. 9.6)

Let R be a commutative ring and $I \trianglelefteq R$. Then $(R/I, +, \cdot)$ is a ring.

Proof. From Lecture 22, $(R/I, +)$ is an abelian group. We must verify that multiplication is well-defined, associative, has an identity, and is distributive over addition.

Well-definedness. Suppose $a_1 + I = a_2 + I$ and $b_1 + I = b_2 + I$. Then $a_1 - a_2 \in I$ and $b_1 - b_2 \in I$. We want

$$(a_1 + I)(b_1 + I) = (a_2 + I)(b_2 + I), \quad \text{i.e.,} \quad a_1b_1 + I = a_2b_2 + I,$$

which means $a_1b_1 - a_2b_2 \in I$. Compute

$$a_1b_1 - a_2b_2 = a_1b_1 - a_1b_2 + a_1b_2 - a_2b_2 = a_1(b_1 - b_2) + (a_1 - a_2)b_2.$$

By the absorption property, $a_1(b_1 - b_2) \in I$ and $(a_1 - a_2)b_2 \in I$. Hence the sum lies in I , so multiplication is well-defined.

Associativity.

$$[(a + I)(b + I)](c + I) = (ab + I)(c + I) = (ab)c + I = a(bc) + I = (a + I)[(b + I)(c + I)].$$

Distributivity.

$$\begin{aligned} (a + I)[(b + I) + (c + I)] &= (a + I)[(b + c) + I] \\ &= a(b + c) + I = (ab + ac) + I \\ &= (ab + I) + (ac + I) = (a + I)(b + I) + (a + I)(c + I). \end{aligned}$$

The right distributive law is analogous.

Multiplicative identity. $1 + I$ satisfies $(1 + I)(a + I) = a + I = (a + I)(1 + I)$.

Therefore R/I is a ring. □

2 Examples of Quotient Rings

Example 2.1: $\mathbb{Z}/n\mathbb{Z} = \mathbb{Z}_n$ (from lecture)

Take $R = \mathbb{Z}$ and $I = n\mathbb{Z}$. Then

$$\mathbb{Z}/n\mathbb{Z} = \{0 + n\mathbb{Z}, 1 + n\mathbb{Z}, \dots, (n-1) + n\mathbb{Z}\} \cong \mathbb{Z}_n,$$

with addition and multiplication being addition and multiplication modulo n . This is the prototype example of a quotient ring.

Example 2.2: $\mathbb{R}[x]/\langle x \rangle$ (from lecture; cf. Lee, Example 9.25)

Take $R = \mathbb{R}[x]$ and $I = \langle x \rangle = x \cdot \mathbb{R}[x]$. By definition, every element of I is of the form

$$g(x) \cdot x = a_0x + a_1x^2 + \dots + a_nx^{n+1}, \quad a_i \in \mathbb{R},$$

i.e., I consists of all polynomials with *no constant term*. Then for $f(x) \in \mathbb{R}[x]$ with $f(x) = c + x \cdot h(x)$,

$$f(x) + I = c + I.$$

Hence

$$\mathbb{R}[x]/\langle x \rangle = \{c + I : c \in \mathbb{R}\},$$

and intuitively $\mathbb{R}[x]/\langle x \rangle \cong \mathbb{R}$ (we will make this isomorphism precise shortly via the First Isomorphism Theorem).

Remark 2.1: Non-commutative case (from lecture)

If R is not commutative, one can still define quotient rings, but one needs a *two-sided ideal* I : a subset $I \subseteq R$ that is an additive subgroup and satisfies

$$rI \subseteq I \quad \text{and} \quad Ir \subseteq I \quad \text{for all } r \in R.$$

The well-definedness of multiplication in R/I then carries through exactly as in the proof of Theorem 1.1, since both $a_1(b_1 - b_2)$ and $(a_1 - a_2)b_2$ require absorption *on each side*.

3 Ring Homomorphisms

Definition 3.1: Ring homomorphism (from lecture; cf. Lee, Def. 9.4)

Let R and S be rings. A *ring homomorphism* is a function $\varphi : R \rightarrow S$ such that, for all $r_1, r_2 \in R$:

- (1) $\varphi(r_1 + r_2) = \varphi(r_1) + \varphi(r_2)$,
- (2) $\varphi(r_1 r_2) = \varphi(r_1) \varphi(r_2)$,
- (3) $\varphi(1_R) = 1_S$.

Remark 3.1: On condition (3) (from lecture)

Lee's textbook (Def. 9.4) omits condition (3): in that convention, $\varphi(1_R)$ need not be 1_S , but it is always the multiplicative identity of $\text{Im } \varphi$ (see Proposition 3.1). Our lecturer adopts the stricter convention requiring $\varphi(1_R) = 1_S$.

Definition 3.2: Kernel and image (from lecture; cf. Lee, Def. 9.5)

Let $\varphi : R \rightarrow S$ be a ring homomorphism. The *kernel* is

$$\ker \varphi = \{r \in R : \varphi(r) = 0\},$$

and the *image* is

$$\text{Im } \varphi = \{\varphi(r) : r \in R\} \subseteq S.$$

Lemma 3.1: Basic properties of $\ker$ and Im (from lecture; cf. Lee, Thm. 9.9)

Let $\varphi : R \rightarrow S$ be a ring homomorphism. Then:

- (1) $\ker \varphi$ is a two-sided ideal of R ;
- (2) $\ker \varphi = \{0\}$ if and only if φ is injective;
- (3) $\text{Im } \varphi$ is a subring of S .

Proof. **(1)** Since φ is a group homomorphism under addition, $\ker \varphi$ is an additive subgroup of $(R, +)$. It remains to check the absorption property. For $x \in \ker \varphi$ and $r \in R$,

$$\varphi(rx) = \varphi(r)\varphi(x) = \varphi(r) \cdot 0 = 0,$$

so $rx \in \ker \varphi$. Similarly $xr \in \ker \varphi$. Thus $\ker \varphi$ is a two-sided ideal.

(2) This follows from the corresponding fact for group homomorphisms applied to addition.

(3) $\text{Im } \varphi$ is an additive subgroup since φ is a group homomorphism. For multiplicative closure: take $s_1, s_2 \in \text{Im } \varphi$, say $s_i = \varphi(r_i)$. Then

$$s_1 s_2 = \varphi(r_1) \varphi(r_2) = \varphi(r_1 r_2) \in \text{Im } \varphi.$$

So $\text{Im } \varphi$ is a subring of S . □

Proposition 3.1: $\varphi(1_R)$ is the identity of $\text{Im } \varphi$ (from lecture)

If we drop condition (3) in Definition 3.1, then $\varphi(1_R)$ is *not necessarily* the multiplicative identity of S . However, it is always the multiplicative identity of $\text{Im } \varphi$.

Proof. Let $s \in \text{Im } \varphi$, say $s = \varphi(r)$. Then

$$\varphi(1_R) \cdot s = \varphi(1_R)\varphi(r) = \varphi(1_R \cdot r) = \varphi(r) = s,$$

and similarly $s \cdot \varphi(1_R) = s$. So $\varphi(1_R)$ acts as the identity on $\text{Im } \varphi$. □

4 Examples and Cautions

Example 4.1: First-factor projection (from lecture)

Define $\varphi : R \rightarrow R \times R$ by $\varphi(r) = (r, 0)$. Then

$$\varphi(r_1 + r_2) = (r_1 + r_2, 0) = (r_1, 0) + (r_2, 0) = \varphi(r_1) + \varphi(r_2),$$

$$\varphi(r_1 r_2) = (r_1 r_2, 0) = (r_1, 0)(r_2, 0) = \varphi(r_1)\varphi(r_2).$$

So φ preserves both operations. But the multiplicative identity of $R \times R$ is $(1, 1)$, whereas

$$\varphi(1) = (1, 0) \neq (1, 1).$$

Hence φ satisfies (1) and (2) but *fails* condition (3); under the strict convention, φ is not a ring homomorphism. Note however that $(1, 0)$ *is* the identity of $\text{Im } \varphi = R \times \{0\}$, consistent with Proposition 3.1.

Example 4.2: $\text{Im } \varphi$ need not be an ideal of S (from lecture)

Consider the inclusion $\varphi : \mathbb{Z} \hookrightarrow \mathbb{Q}$, $\varphi(n) = n$. This is a (injective) ring homomorphism, and

$$\text{Im } \varphi = \mathbb{Z} \subseteq \mathbb{Q}.$$

But $\mathbb{Z}$ is *not* an ideal of $\mathbb{Q}$: take $\frac{1}{2} \in \mathbb{Q}$ and $1 \in \mathbb{Z}$, then

$$\frac{1}{2} \cdot 1 = \frac{1}{2} \notin \mathbb{Z}.$$

So absorption fails. This shows that, in general, $\text{Im } \varphi$ is only guaranteed to be a *subring* of S , not an ideal.

Example 4.3: Reduction modulo n (from Lee, Example 9.12)

For $n \geq 2$, the map $\varphi : \mathbb{Z} \rightarrow \mathbb{Z}_n$ given by $\varphi(a) = [a]$ is a ring homomorphism with $\ker \varphi = n\mathbb{Z}$ and $\text{Im } \varphi = \mathbb{Z}_n$. This is the homomorphism that gives rise (via the First Isomorphism Theorem) to the isomorphism $\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n$ of Example 2.1.

Example 4.4: Complex conjugation (from Lee, Example 9.13)

The map $\varphi : \mathbb{C} \rightarrow \mathbb{C}$, $\varphi(a + bi) = a - bi$, is a ring homomorphism with $\ker \varphi = \{0\}$ (so it is injective) and $\text{Im } \varphi = \mathbb{C}$ (so it is surjective). Thus complex conjugation is a ring automorphism of $\mathbb{C}$.

Checklist of Skills

- ☐ State the definition of the quotient ring R/I and its operations.
- ☐ Prove that multiplication in R/I is well-defined, using the absorption property of I .
- ☐ Verify that R/I is a ring (associativity, distributivity, identity).
- ☐ Identify $\mathbb{Z}/n\mathbb{Z}$ with $\mathbb{Z}_n$ and $\mathbb{R}[x]/\langle x \rangle$ with $\mathbb{R}$.
- ☐ Explain why a two-sided ideal is needed for quotients of non-commutative rings.
- ☐ State the three defining conditions of a ring homomorphism.
- ☐ Compute $\ker \varphi$ and $\operatorname{Im} \varphi$ for given homomorphisms.
- ☐ Prove that $\ker \varphi$ is a two-sided ideal of R .
- ☐ Prove that $\operatorname{Im} \varphi$ is a subring of S .
- ☐ Give an example where $\operatorname{Im} \varphi$ is not an ideal of S (e.g., $\mathbb{Z} \hookrightarrow \mathbb{Q}$).
- ☐ Recognise when $\varphi(1_R) \neq 1_S$, and identify $\varphi(1_R)$ as the identity of $\operatorname{Im} \varphi$.